

NFA TO DFA CONVERSION & CHALLENGES

The Solution & Challenges

- **Conversion Method:** We use the Subset Construction Algorithm (or Power Set Construction).
- **State Explosion Challenge:** An NFA with n states can result in a DFA with up to 2^n states.
- **Resource Cost:** This exponential growth requires high memory and computational power, making it challenging for large machines.



MATHEMATICAL INDUCTION IN AUTOMATA

Role in Automata: Used to rigorously prove an automaton accepts its intended language (e.g., proving string properties by inducting on string length).

Proof for Sum of Squares Formula

Base Case (n=1): The case holds true.
$$1^2 = \frac{1(2)(3)}{6} = 1$$

Inductive Hypothesis: Assume true for an integer k.

Inductive Step (n=k+1): Adding to the hypothesis formula algebraically simplifies to $\frac{(k+1)(k+2)(k+3)}{6}$. This perfectly matches the formula for n=k+1, proving the statement for all positive integers.

$$\frac{(k+1)(k+2)(k+3)}{6}$$

ROLE OF E-TRANSITIONS & NFA CONVERSION

Utility and Conversion

- **Role:** They act as "free moves," allowing the automaton to change states without consuming any input symbols, which simplifies complex machine design.
- **Conversion Step 1:** Compute the ϵ -closure for all states.
- **Conversion Step 2:** Define new transitions using the computed ϵ -closures.
- **Conversion Step 3:** Make a state a final state if its ϵ -closure contains an original final state.



E-CLOSURE & DFA CONSTRUCTION

ϵ -Closure: The set of all states that can be reached from a specific state by following *only* ϵ -transitions.

Step 1: Initial State

The new DFA's start state is defined as the ϵ -closure of the original NFA's start state.

Step 2: Transitions

For an input symbol, find all reachable NFA states from the current DFA state set, then take the ϵ -closure of that result to form the new DFA state.

Step 3: Final States

Any newly created DFA state that contains at least one original NFA final state becomes a final state in the DFA.